

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2020

Bachelor in Computer Applications

Course Title: **C Programming**

Code No: **CACS151** Semester: **II**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is computer programming? Differentiate between top down and bottom up approach of programming.
3. What is the purpose of size of operator? Write a C program to swap the value of two integers without using third variable.
4. Write a program in C to find all possible roots of quadratic equation $ax^2 + bx + c = 0$.
5. Why an array is required in programming? Write a C program to input age of 30 employees into array and find out second largest age from array.
6. What is recursive function? Write C program to generate the Fibonacci series up to 13th term using recursive function.
7. What is the advantage of pointer? Write a C program to store 10 integers into memory and find out minimum and maximum using dynamic memory allocation (DMA).
8. What is the use of `initgraph()` in C program? Write a program to draw line and circle and rectangle using graphic function.

Group C

Attempt any TWO questions[2x10=20]

9. What is structure? Write a C program to enter id, name and address of 25 structure variable called employee and sort them in ascending order on the basis with use of pointer.
10. Is break keyword mandatory in switch statement? Write a program to calculate arithmetic operations (+, -, *, /) using switch case statement.
11. What do you mean by Software Development Life Cycle? Explain waterfall modelw advantage and disadvantage.